

Taylor polynomials and remainder - 6.3

Taylor series for $f(x)$ at a : $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$

Said, e.g. e^x has $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$ as Taylor series.

$\sin(x)$ has predictable derivatives:

$\sin(x), \cos(x), -\sin(x), -\cos(x), \sin(x), \dots$

at 0: $0, 1, 0, -1, 0, \dots$

Taylor series at $a=0$: $\frac{0}{0!} + \frac{1}{1!}x + \frac{0}{2!}x^2 - \frac{1}{3!}x^3 + \frac{0}{4!}x^4 + \dots$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

Taylor series for $\sin(x)$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$,

similarly, can deduce that the Taylor series for $\cos(x)$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

Definition: The k^{th} Taylor Polynomial of f at $x=a$ is the polynomial $T_k(x) = \sum_{n=0}^k \frac{f^{(n)}(a)}{n!} (x-a)^n$.

Note: $T_0(x) = \sum_{n=0}^0 \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a)$ is constant.

$$T_1(x) = f(a) + \frac{f'(a)}{1!} (x-a) = f(a) + f'(a)(x-a) \text{ is}$$

the tangent line! This is the best linear approximation to f near $x=a$.

The polynomial $T_2(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$
is the best quadratic approximation to f near a .
Generally, $T_k(x)$ is the best polynomial
approximation to f near a of degree k .

Example: $T_2(x)$ for $\sin(x)$ near 0 is x .

($T_2(x) = x$ is linear, but still the best
quadratic approximation to $\sin(x)$ near 0.)

Definition: The k^{th} remainder is

$$R_k(x) = f(x) - T_k(x).$$

To say that $f(x)$ equals its Taylor series is
to say that $R_k(x)$ goes to 0 as $k \rightarrow \infty$.

Taylor's Theorem with Remainder: If f can be
differentiated at least $k+1$ times on some
interval I , then for every x in I , there
exists a number c between a and x such
that
$$R_k(x) = \frac{f^{(k+1)}(c)}{(k+1)!} (x-a)^{k+1}.$$

Note that $k=0$ says $f(x) - f(a) = R_0(x) = \frac{f'(c)}{1!} (x-a)$
i.e. $\frac{f(x) - f(a)}{x-a} = f'(c)$ — this is the Mean Value
Theorem.

If we happen to know that for $x \in I$,
 $|f^{(k+1)}(x)| \leq M$, then the theorem says that

$$|R_k(x)| \leq \frac{M}{(k+1)!} |x-a|^{k+1}$$

Example: if $f(x) = \sin(x)$, then $f^{(k+1)}(x)$ is

$\pm \sin(x)$ or $\pm \cos(x)$ or something, so $|f^{(k+1)}(x)| \leq 1$

$$\text{Thus } |R_k(x)| \leq \frac{1}{(k+1)!} |x-a|^{k+1} = \frac{|x|^{k+1}}{(k+1)!}$$

for the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$

Hence $\lim_{k \rightarrow \infty} |R_k(x)| \leq \lim_{k \rightarrow \infty} \frac{|x|^{k+1}}{(k+1)!} = 0$ because factorials grow faster than exponentials.

Thus $\lim_{k \rightarrow \infty} T_k(x) = \sin(x)$, that is, $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$.

Definition: A function f is called analytic if it is equal to its Taylor series on the interval of convergence for the series.

Examples: $\sin(x)$ is analytic, and we showed that the radius of convergence is ∞ . That is, the interval of convergence is $(-\infty, \infty)$. The same is true of $\cos(x)$ and e^x .

Previously, we saw series for $\ln(1+x)$ and $\ln(x)$, so these are also analytic.

Example: Find a power series for $x \cos(x)$
and for $\sin(x^2)$.

$$x \cos(x) = x \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+1}$$

$$\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (x^2)^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+2}$$